

HAMZA KHAN

Graduate Data Engineer · Python · SQL · ETL Pipelines · Cloud

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFESSIONAL PROFILE

Computer science graduate (B.E., Data Science specialisation, CGPI 8.19/10) with four years of freelance delivery experience building Python back-end systems, REST APIs and data pipelines for clients in India and Germany. Recent work has moved deliberately towards data engineering: ETL pipelines loading into PostgreSQL, a local-first analytics engine on DuckDB and FastAPI covering schema detection, cleaning, PII masking and data-quality scoring, and SQL analytics surfaced through Power BI. Comfortable owning delivery end to end — requirements, data modelling, implementation, deployment and support — having worked solo on every engagement to date. Smart India Hackathon 2025 Grand Finalist. Seeking a graduate or junior data engineering role where I can build on strong SQL and Python foundations and grow towards a cloud data engineering specialisation.

KEY SKILLS

Data engineering: ETL and ELT pipeline design, data modelling, schema auto-detection and column mapping, data cleaning and validation, de-duplication, incremental loads, data-quality scoring, PII masking (SHA-256), audit and lineage tracking

Languages: Python, SQL, JavaScript, TypeScript, Bash

Databases: PostgreSQL, MySQL, SQLite, DuckDB, Elasticsearch

Analytics and BI: pandas, NumPy, SQLAlchemy, analytical SQL (CTEs, window functions, partitioned ranking), Power BI, Prophet forecasting, RFM and cohort segmentation, anomaly detection

Cloud and DevOps: Azure, Docker, Git, GitHub Actions (CI), DigitalOcean, Render, Vercel, Linux

Back-end engineering: Django, Django REST Framework, Django Channels (WebSockets), FastAPI, Flask, REST API design, JWT authentication, role-based access control

Quality and practice: pytest, Ruff, automated CI pipelines, technical documentation, version control workflows

Front-end (supporting): React, Next.js, Vite, Tailwind CSS, shadcn/ui, TanStack Query

PROFESSIONAL EXPERIENCE

Full-Stack Developer (Freelance) — Brandenbed Living Spaces, Berlin, Germany (Remote)

September 2025 – November 2025

- Delivered the company's production website for serviced-apartment operations across European markets, achieving a Google PageSpeed Insights score above 95 on desktop.
- Designed and built an internal CRM with role-based access, supporting property management and client communication workflows for the operations team.
- Worked asynchronously across time zones with the Berlin founding team, translating business requirements into shipped features and maintaining written specifications so decisions survived the handover.

Back-End Developer (Freelance) — Self-Employed, Mumbai, India (Remote)

2021 – 2025

- Delivered Django web applications and REST APIs for clients in India and overseas, owning the full lifecycle solo: requirements gathering, database design, development, deployment and ongoing maintenance.
- Built and launched a property-listing marketplace for a MahaRERA-registered channel partner on Django 5.2 and PostgreSQL, with OAuth authentication, a Grappelli administration interface, and property search and filtering. (The client has since replaced the front end with a static build; the original application was database-backed.)
- Built and launched a corporate site for an import/export business on Next.js 15, with a reCAPTCHA-protected enquiry form writing to Google Sheets and automated email notification via Nodemailer.
- Implemented role-based access control and real-time WebSocket features with Django Channels across PostgreSQL-backed CRM and e-commerce platforms.

- Deployed and maintained applications on AWS EC2, DigitalOcean, Render and Hostinger using Docker, including production troubleshooting and direct client support.

DATA AND TECHNICAL PROJECTS

NOVEM — E-Commerce Intelligence Platform | [GitHub](#)

Python · FastAPI · DuckDB · SQLite · pandas · Prophet · React 19 · TypeScript · Tauri

- Built a local-first desktop analytics application that turns e-commerce exports (CSV, Excel, Google Sheets, Shopify Admin API, PostgreSQL) into KPIs, customer segments, forecasts and plain-language insights, with all processing performed on the user's own machine and no data uploaded.
- Designed the ingestion pipeline end to end: schema auto-detection, column mapping, type coercion and cleaning, de-duplication, SHA-256 PII masking and data-quality scoring, all applied before any record is merged into the analytical store.
- Architected a FastAPI analytics engine of 23 routers and roughly 170 endpoints, using DuckDB for columnar analytical tables and SQLite for settings and encrypted credentials.
- Delivered analytics modules including RFM segmentation, churn risk and customer lifetime value (BG/NBD and Gamma-Gamma), cohort retention, market-basket association rules, Prophet forecasting, anomaly detection and SHAP explanations.
- Set up a GitHub Actions CI pipeline running Ruff and pytest on Python 3.11 and 3.12 alongside TypeScript type-checking, ESLint at zero warnings and Vitest, with a PyInstaller/NSIS script producing a Windows installer.

Customer Shopping Behaviour Analysis — ETL and BI Pipeline | [GitHub](#)

Python · pandas · SQLAlchemy · PostgreSQL · SQL · Power BI

- Built an end-to-end analytics pipeline over a retail dataset: profiling and cleaning in pandas, category-level median imputation of missing ratings, derived age-quartile and purchase-cadence fields, load into PostgreSQL via SQLAlchemy, ten business questions answered in SQL, and a Power BI dashboard on top.
- Wrote analytical SQL using CTEs and RANK() OVER (PARTITION BY category) to produce top-items-per-category rankings and segment revenue breakdowns.
- Reported findings honestly, including the conclusion that most customer segments in the dataset do not separate meaningfully — documenting the limits of the data rather than overstating the result.

InitCore CRM — Call Centre Platform | [GitHub](#)

Python · Django 5.0 · PostgreSQL · Django Channels · Daphne · pandas · pdfkit

- Built a production-style CRM for outbound call-centre operations across 17 data models, covering lead management, call dispositions, payment verification, GST invoicing, attendance and complaint tracking.
- Implemented CSV and XLSX lead ingestion with interactive column mapping, de-duplication against existing records and full audit history — the same import-and-reconcile problem that sits at the centre of most data engineering work.
- Implemented three-tier role-based access (administrator, team leader, agent) enforced at queryset level, so a single set of views serves all roles while strictly scoping the data each one can see.
- Delivered live agent break monitoring over Django Channels WebSockets, and automated GST invoice generation rendered to PDF.

FlaskSearch API — Full-Text Search Service | [GitHub](#)

Python · Flask · Elasticsearch · Docker · REST

- Built a Flask service exposing full-text search over a Shakespeare plays corpus, running multi_match queries across title, author and character fields against an Elasticsearch index, with REST endpoints for listing, searching and indexing documents and Elasticsearch containerised in Docker for reproducible local setup.

Conference Websites — SUSTECH 2025 and ICTMT 2025 | [SUSTECH](#) | [ICTMT](#)

React 19 · Vite 6 · Tailwind CSS 4 · shadcn/ui · Framer Motion

- Delivered two production single-page conference sites for Saraswati College of Engineering covering tracks, key dates, calls for papers, fee tables and committee listings, with all copy held in a single data module so organisers could update deadlines and committees without touching markup.

Developer Portfolio | [Live](#) | [GitHub](#)

React 18 · TypeScript · Vite · Tailwind CSS · shadcn/ui

- Type-safe responsive personal site with dark mode and motion transitions, optimised for Core Web Vitals and accessibility.

EDUCATION

Bachelor of Engineering — Computer Science and Engineering (Data Science)

Saraswati College of Engineering, Kharghar (University of Mumbai) | September 2023 – July 2026 | CGPI 8.19/10 (73.73%)

- Entered by lateral admission into second year on the strength of the diploma. Strongest semesters: V (SGPI 9.32), VIII (8.71), VII (8.50).
- Data-relevant coursework: Data Warehousing and Mining, Big Data Analytics, Data Analytics and Visualisation, Statistics for AI and Data Science, Machine Learning, Deep Learning, Distributed Computing, Cloud Computing, DBMS, Data Structures, Analysis of Algorithms, Computer Networks, Operating Systems, Cryptography and System Security, Software Engineering and Project Management.

Diploma in Computer Engineering — First Class

A. R. Kalsekar Polytechnic, New Panvel (Maharashtra State Board of Technical Education) | 2021 – 2023 | Aggregate 72.63%

- Three-year technical diploma covering data structures, DBMS, operating systems, networks, Java, Python, PHP, software testing and mobile app development, plus supervised industrial training and a capstone project.

Higher Secondary Certificate (Science) — Maharashtra State Board

2021 | 87.50% | Physics, Chemistry, Mathematics and Statistics, Information Technology, English

Secondary School Certificate — Maharashtra State Board

2019 | 81.20% (Distinction)

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Complete Data Analyst Bootcamp: From Basics to Advanced — Udemy (KRISHAI Technologies), February 2026. Python, pandas, NumPy, SQL, exploratory data analysis, statistics, data visualisation.
- The Complete Full-Stack Web Development Bootcamp — Udemy (Dr Angela Yu), February 2026. Node.js, Express.js, MongoDB, REST APIs, Git.
- Excel Basics for Data Analysis and Introduction to Data Analytics — Coursera (IBM), December 2025 and January 2026.
- Introduction to Data Science — Great Learning, October 2025. Aptitude, Life Skills and Technical Training Programme, November 2025.

AWARDS AND ACHIEVEMENTS

Smart India Hackathon 2025 — Grand Finalist

Ministry of Education Innovation Cell and AICTE, Government of India | Biju Patnaik University of Technology, Rourkela | December 2025

- Selected for the national Grand Finale from over 50,000 competing teams; led a team through a 36-hour sprint to a working solution.

SCOE Avishkar 2025 — Second Place

Project competition, Saraswati College of Engineering (University of Mumbai) | April 2025

- Led a team to second place in a university-level engineering project competition.

CodeCrafters 2.0 Hackathon — Participant

University of Mumbai | March 2025

- Built a digital asset management tool for financial instruments using Spring Boot and React within a 48-hour sprint.

ADDITIONAL INFORMATION

Languages: English (professional working proficiency — IELTS Academic 6.5, June 2026), Hindi (native), Marathi (conversational)

Interests: Data engineering, cloud data platforms, back-end systems and open source. Eight public repositories, each with full technical documentation.